ASCOMP 2026 POSTER

Min-Marginal Power Indices for Multi-Objective Feature Selection

Wenjie Shi & Chi Zhao*
Saint Petersburg State University

7/9 Universitetskaya nab., Saint Petersburg, 199034, Russia
st147487@student.spbu.ru
* Corresponding author: chi.zhao@spbu.ru

OPTIMIZATION OBJECTIVES — **Minimize feature count + Maximize predictive utility**

01 CORE IDEA & METHOD

From average importance to worst-case evidence.
MinPI detects feature additions that push utility below the quota.

Acceptable	+ feature i	Unacceptable
v(S) = 1	marginal = -1	v(S ∪ {i}) = 0

THE WORST-CASE MARGINAL

$$v_\alpha(S) = 1[U(M_S) \geq \alpha\, U_{full}]$$

$$\mathbf{MinPI_i = \min [v_\alpha(S \cup \{i\}) - v_\alpha(S)]}$$

over $S \subseteq N \setminus \{i\}$; N = analyzed features

Validation-gated feature removal

H: features with an evaluated harmful coalition witness (-1).

$$\mathbf{U_{val}(N \setminus H) \geq U_{val}(N) - \varepsilon}$$

ε = 0.005 (classification); 0.01 (regression).

Pass: remove H. Fail: remove nothing. Models fit on training data.

SCAN ME

CORE INSIGHT

From harmful-coalition evidence to validation-gated feature removal.

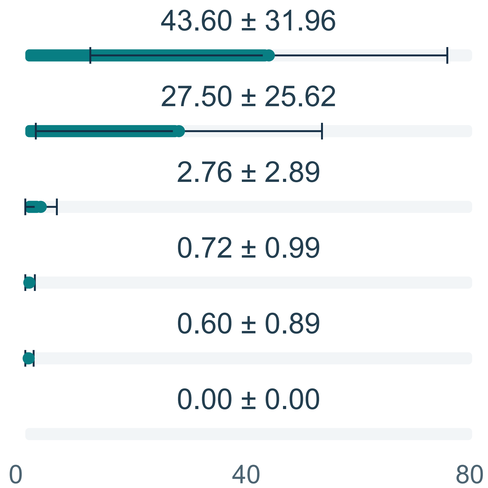
Scan to read the full paper
hlnks.co/63671101

02 EXPERIMENTS & RESULTS

6 datasets × 5 seeds | XGBoost | α = 0.95
64/16/20% train/validation/test; ≤100 analyzed features.

Five-seed mean ± SD; removal of analyzed features.
ΔU = held-out utility - full-model utility.

DATASET	REMOVED (%)	HELD-OUT ΔU
Madelon	43.60 ± 31.96	+0.00469 ± 0.00497
Abalone	27.50 ± 25.62	-0.00565 ± 0.01077
Credit-Card-Fraud	2.76 ± 2.89	-0.00078 ± 0.00335
Spambase	0.72 ± 0.99	-0.00003 ± 0.00007
Gisette	0.60 ± 0.89	-0.00011 ± 0.00016
ISOLET	0.00 ± 0.00	0.00000 ± 0.00000



12.53% mean feature removal

-0.000313 mean held-out ΔU

NSGA-II: mean held-out ΔHV +0.01584; 800 evaluations/variant.
Madelon: +0.09515. Overall Holm p = 0.4375.

MADELON: MATCHED-COUNT REMOVAL

MinPI ΔU +0.0047 | Random ΔU -0.0732